

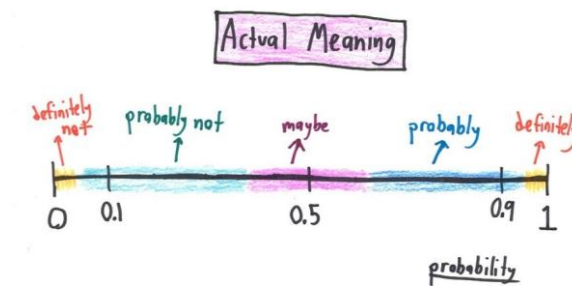


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 8 - Lecture No 15 – March 02, 2026



Announcements!

1. Office hours (Ramadan) – **Tuesday & Thursday 11:30 to 12:30 PM** @ Room C-211 (or through meeting request)
2. **Midterm Examination – March 3, 2026, from 03:25 PM to 04:55 PM at the Tariq Rafi Lecture Hall**
3. Quiz No 05 – March 18, 2026
4. Assignment 2 – due March 8, 2026
5. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE – Spring 2026							
Course	Tutor	Email	Monday	Tuesday	Wednesday	Thursday	Friday
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	2:00-3:00		2:00-3:00		12:40-1:40
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	12:45-2:30		12:45-2:30		11:00-12:00
	Zain Naqi	zn09224@st.habib.edu.pk		9:00 - 10:30	11:45 - 12:45	9:00 - 10:30	

Agenda for today

1. Quiz No 04 (on DRV)

Unit 4
Discrete Random Variables

Multiple random variables and joint PMFs

- $X : p_X$
 - $Y : p_Y$
- marginal pmfs*
- $P(X = Y) = 2/20$

Joint PMF: $p_{X,Y}(x, y) = P(X = x \text{ and } Y = y)$

Y =
Second roll

4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll $P_X(3)$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$P_{X,Y}(1, 3) = 2/20$$

$$P_X(4) = 1/20 + 2/20$$

$$P_Y(2) = (1 + 3 + 1)/20$$

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$



Exercise No 01!

Suppose you roll two fair, six-sided dice, one of which is **red** and the other of which is **green**. Define the following two random variables **X** & **Y**:

X = The number shown on the **red die**

Y =
$$\begin{cases} 0 & \text{if the two dice show the same number} \\ 1 & \text{if the number on the green die is bigger than the number on the red die} \\ 2 & \text{if the number on the red die is bigger than the number on the green die} \end{cases}$$

Example



X =
Y =

1. Write down a table showing the joint probability mass function for **X** and **Y**.

		X =					
		1	2	3	4	5	6
Y =	0						
	1						
	2						



Exercise No 01!

$X =$

		1	2	3	4	5	6
	0	1/36	1/36	1/36	1/36	1/36	1/36
$Y =$	1	5/36	4/36	3/36	2/36	1/36	0
	2	0	1/36	2/36	3/36	4/36	5/36

2. Find the marginal probability mass function for Y , and compute its expected value.

y	$P(Y = y)$
0	
1	
2	

$$E(Y) = 0 \times (6/36) + 1 \times (15/36) + 2 \times (15/36) = 45/36$$



Exercise No 01!

$X =$

		1	2	3	4	5	6
$Y =$	0	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
	1	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$	0
	2	0	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$

3. Find the conditional probability mass function for X given $Y = 1$.

$X =$

1	2	3	4	5	6
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More than two random variables

$$p_{X,Y,Z}(x, y, z) = \mathbf{P} (X = x \text{ and } Y = y \text{ and } Z = z)$$

$$\rightarrow \sum_x \sum_y \sum_z p_{X,Y,Z}(x, y, z) = 1$$

$$\rightarrow p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x, y, z)$$

$$\rightarrow p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z)$$

Functions of multiple random variables

$$Z = g(X, Y)$$

$$\rightarrow p_Z(z) = \mathbf{P}(Z = z) = \mathbf{P}(g(X, Y) = z) = \sum_{(x, y): g(x, y) = z} p_{X, Y}(x, y)$$

Expected value rule:

$$\rightarrow E[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X, Y}(x, y)$$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$\rightarrow E[X+Y]=E[g(x, y)]$$

$$\rightarrow g(x, y)=x+y$$

$$\rightarrow = \sum_x \sum_y (x+y) p_{X,Y}(x,y)$$

$$\rightarrow = \sum_x \sum_y (x) p_{X,Y}(x,y) + \sum_x \sum_y (y) p_{X,Y}(x,y)$$

$$\rightarrow = \sum_x x \sum_y p_{X,Y}(x,y) + \sum_y y \sum_x p_{X,Y}(x,y)$$

$$\rightarrow = \sum_x x p_X(x) + \sum_y y p_Y(y) = E[X] + E[Y]$$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$$

$$E[2X + 3Y - Z] = E[2X] + E[3Y] - E[Z] = 2E[X] + 3E[Y] - E[Z]$$

Conditional PMFs

$$A = \{ Y = y \}$$

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A)$$

$$p_{X|Y}(x \mid y) = \mathbf{P}(X = x \mid Y = y) = \frac{p(X = x, Y = y)}{p(Y = y)}$$

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \quad \text{defined for } y \text{ such that } p_Y(y) > 0$$

$$\sum_x p_{X|Y}(x|y) = 1$$

Y = Second roll

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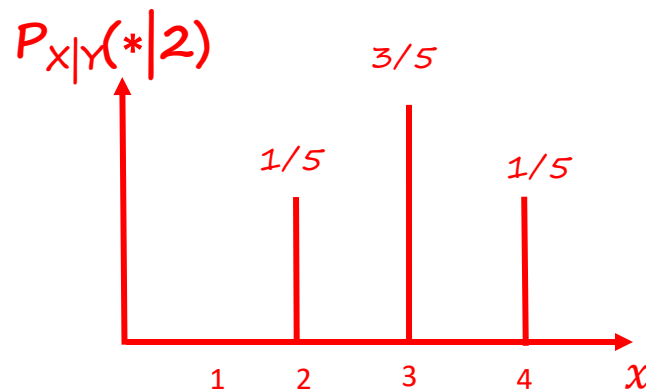
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll

$$\rightarrow P_Y(2) = 5/20$$

$$\rightarrow P_{X|Y}(1|2) = 0$$

$$\rightarrow P_{X|Y}(2|2) =$$



$$p_{X,Y}(x, y) = p_Y(y)p_{X|Y}(x \mid y)$$

$$p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y \mid x)$$

Conditional PMFs involving more than two r.v.'s

- **Self-explanatory notation**

$$\begin{aligned}\rightarrow p_{X|Y,Z}(x|y,z) &= p(X=x \mid Y=y, Z=z) = \frac{p(X=x, Y=y, Z=z)}{p(Y=y, Z=z)} \\ &\rightarrow = \frac{p_{X,Y,Z}(x, y, z)}{p_{Y,Z}(y, z)}\end{aligned}$$

$$\rightarrow p_{X,Y|Z}(x, y|z) = p(X=x, Y=y \mid Z=z)$$

- **Multiplication rule**

$$P(A \cap B \cap C) = P(A) P(B \mid A) P(C \mid A \cap B)$$

$$A = \{X=x\} \quad B = \{Y=y\} \quad C = \{Z=z\}$$

$$p_{X,Y,Z}(x, y, z) = p_X(x) p_{Y|X}(y \mid x) p_{Z|X,Y}(z \mid x, y)$$